Measuring Sentiment Trends Under Model and Processing Constraints

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Abstract - Sentiment monitoring often converts post-level labels into aggregate trends by classifying and counting. This study evaluates the resulting measurement risk alongside model accuracy and local processing constraints. VADER, TF-IDF logistic regression, an existing BiLSTM and a Twitter RoBERTa checkpoint are compared on 12,284 labeled TweetEval posts. RoBERTa achieves 0.722 accuracy and 0.724 macro F1. A split-calibrated three-class quantification diagnostic evaluates simple counting and adjusted counting on 1,980 synthetic prevalence bags at n=500. Adjusted counting reduces prevalence mean absolute error from 0.099 to 0.026 for RoBERTa and from 0.158 to 0.045 for VADER. On 127,741 historical Bluesky posts, VADER and RoBERTa retain opposite aggregate net-sentiment signs under duplicate, coverage and author-weighting sensitivity checks. A controlled replay further shows that higher-cost inference can reduce completed records under a bounded queue. The study does not claim Bluesky accuracy because no independent target-domain labels are available.

Index Terms - *sentiment quantification, Bluesky, social media monitoring, RoBERTa, reproducibility*

I. INTRODUCTION

Sentiment dashboards are usually used to estimate an aggregate property of a corpus rather than to inspect a single post. The common classify-and-count approach therefore turns classification errors into potential trend error. This project asks whether practical model selection changes aggregate sentiment, whether a calibration-based correction changes the result on held-out labeled data, and whether inference cost affects which offered posts reach storage.

The contribution is an empirical, reproducible diagnostic rather than a new classifier. It combines (i) a common labeled benchmark, (ii) a disjoint split-calibrated quantification evaluation, (iii) historical Bluesky sensitivity analysis, and (iv) a bounded-queue replay with actual SQLite insertion. The paper distinguishes benchmark correctness, aggregate estimation error and operational behaviour.

II. RELATED WORK

TweetEval provides a standardized Twitter classification benchmark [1]. Twitter-specific language models and the Cardiff NLP checkpoint used here account for temporal and domain adaptation in social-media language [2], [3]. VADER supplies a transparent lexicon baseline [4], while bidirectional recurrent models provide the basis for the existing BiLSTM [5], [6].

Sentiment quantification estimates class prevalence in an unlabeled collection. Moreo and Sebastiani show that evaluating only post-level classification can be inadequate when the downstream goal is prevalence estimation [7]. Daughton and Paul study the bias introduced by classify-and-count and uncertainty-aware correction [8]. Existing Bluesky research has demonstrated the value of transparent monitoring systems [9] and the AT Protocol makes public social-media research technically feasible [10]. The present study focuses on how different model outputs and throughput constraints affect a single monitor, without claiming platform-wide opinion measurement.

III. EXPERIMENTAL DESIGN

All four models receive the same 12,284 TweetEval test rows. VADER applies compound thresholds of -0.05 and +0.05. The learned baseline is TF-IDF over unigrams and bigrams with logistic regression trained only on 45,615 local training rows. The BiLSTM uses row-checked saved predictions because fresh TensorFlow inference is unavailable on this host. RoBERTa uses cardiffnlp/twitter-roberta-base-sentiment-latest at a cached revision, original local text, 128-token truncation and CPU inference.

For the quantification diagnostic, the labeled test rows are stratified into disjoint halves of 6,141 and 6,143 rows with seed 42. One half estimates each model's predicted-by-true three-class confusion matrix. The other half supplies 66 prevalence settings, each repeated 30 times at bag sizes 100, 500 and 1,000. Samples are drawn with replacement within each true class, and every model receives the same bags. Classify-and-count uses prediction shares. Adjusted classify-and-count estimates the non-negative class-prevalence vector that best explains observed prediction shares under the calibration matrix. The primary result uses n=500; the direction is reversed at n=500 as a sensitivity check. This is a diagnostic on an already inspected benchmark, not an independent benchmark or a correction asserted for Bluesky.

The historical corpus contains 127,741 cleaned English posts returned for the literal query AI between 19 October and 1 November 2026 UTC. A search response cap can truncate a window. We therefore report duplicate-text, uncapped-hour, literal-token and equal-author sensitivity variants. Net sentiment is the mean of label scores: negative=-1, neutral=0 and positive=1. No historical post labels are used as ground truth.

TABLE I. COMMON LABELED TEST PERFORMANCE

Model	Accuracy	Macro F1	Macro recall
VADER	0.530	0.529	0.570
TF-IDF + LR	0.592	0.560	0.567
BiLSTM	0.595	0.558	0.558
RoBERTa	0.722	0.724	0.735

Scores use the same 12,284 TweetEval test posts. BiLSTM values come from validated saved predictions.

IV. RESULTS

RoBERTa obtains the strongest observed benchmark result, with 0.722 accuracy and 0.724 macro F1. The linear baseline and saved BiLSTM scores are close on this split, so their apparent ordering should not be treated as an architecture-only finding. RoBERTa's superior post-level result motivates testing whether its advantage remains meaningful for the aggregate task.

TABLE II. PRIMARY QUANTIFICATION DIAGNOSTIC AT BAG SIZE 500

Model	CC P-MAE	ACC P-MAE	CC sign err.	ACC sign err.
VADER	0.158	0.045	22.9%	1.3%
TF-IDF + LR	0.193	0.044	19.4%	1.3%
BiLSTM	0.211	0.048	13.6%	1.4%
RoBERTa	0.099	0.026	5.8%	0.1%

P-MAE is mean absolute error across negative, neutral and positive prevalence. Sign error excludes true net sentiment within +/-0.1; 1,800 bags are eligible.

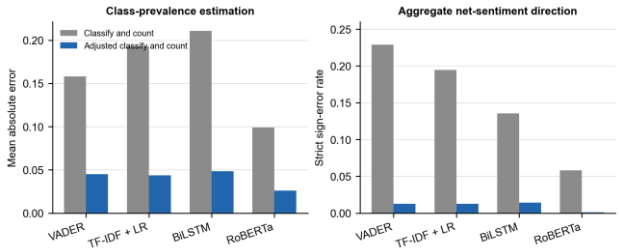


Fig. 1. Disjoint split-calibrated quantification diagnostic at n=500. Adjusted counting improves each observed model under this evaluation protocol.

At n=500, simple counting has prevalence mean absolute error of 0.099 for RoBERTa, 0.158 for VADER, 0.193 for the linear baseline and 0.211 for the saved BiLSTM. Adjusted counting reduces these values to 0.026, 0.045, 0.044 and 0.048. RoBERTa's aggregate net-sentiment error falls from 0.123 to 0.034. The reverse split produces the same ordering for adjusted-count prevalence error, with RoBERTa at 0.024 and the remaining models between 0.044 and 0.046. These results show that post-level accuracy alone does not describe downstream trend reliability.

TABLE III. HISTORICAL VADER AND ROBERTA NET-SENTIMENT SENSITIVITY

Variant	Units	VADER	RoBERTa
All posts	127,741	+0.097	-0.248
Unique normalized text	121,111	+0.092	-0.262
Uncapped hours	69,168	+0.099	-0.252
Equal author-day	87,823	+0.079	-0.304

Positive values indicate a positive aggregate net sentiment. The table compares outputs, not target-domain correctness.

The two models give opposite signs in all 14 observed daily means. The daily equal-weight VADER-minus-RoBERTa difference is 0.345, with a day-resampling interval from 0.333 to 0.360. The historical sample is concentrated: its most active 1% of authors produce 24.4% of posts. Yet the sign difference persists when each author-day receives equal weight. This is evidence of model sensitivity, not evidence that one model measures Bluesky sentiment correctly.

The controlled replay offers 288 posts per model-rate condition across three runs. VADER stores all 288 records at 10, 40 and 80 posts/s. RoBERTa stores 191, 79 and 72 records respectively, with the remaining posts dropped when the queue is full. These finite tests include classifier execution and SQLite insertion, but exclude live network delivery, filtering and collector orchestration.

V. DISCUSSION AND LIMITATIONS

The primary finding is not that a particular classifier is universally best. It is that a monitoring system must evaluate the aggregate quantity it reports. Calibration-based adjustment improves aggregate estimates on the labeled source benchmark, while model selection changes the sign and magnitude of the historical signal. A system designer should therefore report classifier identity, calibration domain, collection coverage and operational drop behaviour beside any public trend.

The paper has important limits. The quantification diagnostic samples from TweetEval, so it does not establish that the same correction transfers to Bluesky. The historical search has a single query, a short period, possible cap truncation and unknown API recall. Polarity is not stance toward AI. The BiLSTM's fresh runtime is unavailable and its benchmark values use saved predictions. The replay uses one machine and a finite workload. Most importantly, independent Bluesky labels are absent; no in-domain accuracy, prevalence-error or correction claim is made.

VI. CONCLUSION

This study connects post-level sentiment classification with the aggregate and operational decisions made by a monitor. On a common benchmark, RoBERTa has the highest observed scores. A disjoint split-calibrated diagnostic shows that adjusted counting materially reduces aggregate estimation error for every model studied. Historical Bluesky outputs remain model-sensitive under several aggregation checks, and controlled replay shows that processing constraints influence record completion. The evidence supports transparent, reproducible monitoring claims while retaining clear boundaries on target-domain correctness.

ACKNOWLEDGMENT

OpenAI Codex was used to help draft and edit portions of the Abstract, Introduction, Related Work, Discussion and Conclusion from author-provided experiment records. The author is responsible for the study design, verification, final text and all reported claims.

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